Sparse & Explainable Bayesian Neural Networks for Official Statistics

Unit-Weighted Pseudo-Posteriors, Formal Differential Privacy, and Non-Probability Propensity Estimation in LBBNN

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Section 1

PART I: Foundations of LBBNN & Input-Skip Architecture (ISLaB)

Motivation — Why Sparse & Explainable BNNs in Official Statistics?

Challenges with Standard Deep Learning in Federal Agencies:

- Standard deep neural networks capture complex non-linearities and multi-way interactions, but are massively over-parameterized, black-box, and lack reliable uncertainty quantification.
- Standard dense Bayesian Neural Networks (BNNs) quantify weight uncertainty, but retain all extraneous noise variables, causing severe variance inflation in downstream survey estimators.

Our Core Objectives with `LBBNN` & `ISLaB`:

1. **Joint Structural & Parameter Uncertainty**: Unify spike-and-slab topological edge selection with variational weight distributions in a single probabilistic framework.
2. **Dynamic Adaptation (Depth Metric)**: Automatically adapt between simple linear models (Depth $= 1$) and deep non-linear interaction surfaces (Depth ≥ 2).
3. **Federal Statistical Extensions**: Extend `LBBNN` to handle informative survey sampling weights, (ϵ, δ)-differential privacy, and non-probability sample integration.

From Dense BNNs to Latent Binary BNNs (LBBNNs)

Probabilistic Formulation (Hubin & Storvik, 2024):

- Likelihood $y_i \sim f(\cdot \mid \zeta(\mathbf{x}_i))$, where hidden layer pre-activations include binary inclusion indicators:

$$a_p^{(j)} = o^{(j)}\left(w_{0,p}^{(j)} + \sum_{k=1}^{K} \gamma_{k,p}^{(j)} a_k^{(j-1)} w_{k,p}^{(j)}\right), \qquad \gamma_{k,p}^{(j)} \in \{0,1\}$$

- **Spike-and-Slab Structural Prior**:

$$p(w_{k,p}^{(j)} \mid \gamma_{k,p}^{(j)}) = \gamma_{k,p}^{(j)} \mathcal{N}(0, (\tau_{k,p}^{(j)})^2) + (1 - \gamma_{k,p}^{(j)})\delta_0, \qquad p(\gamma_{k,p}^{(j)}) = \text{Bernoulli}(\psi_{k,p}^{(j)})$$

Inference & Dual Operational Modes:

- **Bayesian Model Averaging (BMA)**: Ensembles across posterior draws $\Gamma \sim q(\Gamma)$ and $W \sim q(W \mid \Gamma)$ for optimal uncertainty quantification and predictive calibration.
- **Median Probability Model (MPM)**: Hard-prunes all edges whose posterior inclusion probability $\alpha_{k,p}^{(j)} = \mathbb{P}(\gamma_{k,p}^{(j)} = 1 \mid \mathcal{D}) \leq 0.5$ (Barbieri & Berger, 2004 optimal predictive submodel).

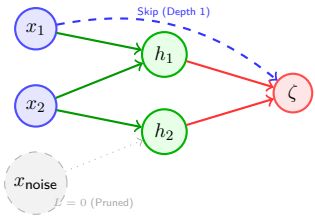
Input-Skip LBBNNs (ISLaB) & Structural Depth Metrics

The ISLaB Architecture (Høyheim et al., 2026, *JAIR*):

- Direct skips route inputs $\mathbf{x} \in \mathbb{R}^p$ to all hidden layers: $\mathbf{a}^{(j-1)} = [a_1^{(j-1)}, \ldots, a_K^{(j-1)}, \mathbf{x}^\top]^\top$.
- **Active Paths (AP)**: Connects input x_k to output ζ if all intermediate edges have $\alpha_{k,p}^{(j)} > 0.5$.

Contribution Depth Classification:

- **Depth = 1**: Input-to-output skip $\implies$ Linear GLM effect.
- **Depth $\geq$ 2**: Through hidden nodes $\implies$ Non-linear/interaction.
- **Zero Paths ($L = 0$)**: Covariate completely pruned.



Local Explainability via Piecewise-Linear Activation Regions

Local Generalized Linear Model Representation (Hubin et al., JSM 2026):

- With continuous piecewise-linear activations (e.g., ReLU), the input space $\mathbb{R}^p$ is partitioned into a finite set of convex polyhedral activation regions $\mathcal{P}_1, \ldots, \mathcal{P}_M$.
- Within any local neighborhood $B_\epsilon(\mathbf{x}_i) \subset \mathcal{P}_m$, the active network collapses to an exact **local Generalized Linear Model (GLM)**:

$$\zeta(\mathbf{x}) = \beta_0(\mathbf{x}_i) + \sum_{j=1}^{p} \beta_j(\mathbf{x}_i) x_j, \qquad \boldsymbol{\beta}(\mathbf{x}_i) = \nabla_{\mathbf{x}} \zeta(\mathbf{x})\big|_{\mathbf{x}=\mathbf{x}_i}$$

Rigorous Statistical Properties for Official Statistics:

- **Closed-Form Gradients**: Local coefficients $\beta_j(\mathbf{x}_i)$ are evaluated exactly via backpropagation without numerical perturbation approximations (e.g., unlike LIME or Kernel SHAP).
- **Bayesian Local Credible Intervals**: Sampling posterior weights $W \sim q(W \mid \Gamma^*)$ yields exact posterior distributions $p(\boldsymbol{\beta}(\mathbf{x}_i) \mid \mathcal{D})$ for local covariate effects on individual survey respondents.

Variational ELBO, Cold-Posterior Protection & Software Ecosystem

Variational Inference Framework:

- Mean-field spike-and-slab: $q(\mathbf{w}, \boldsymbol{\gamma}) = \prod q(w_{k,p}^{(j)} \mid \gamma_{k,p}^{(j)})\, q(\gamma_{k,p}^{(j)})$.
- Efficient sampling via Local Reparameterization Trick (LRT) and Normalizing Flows (*TMLR*, 2024).

Protection Against Cold Posteriors (Wenzel et al., 2020):

- Exact ELBO at $T = 1.0$: $\text{ELBO}(q) = \sum_{i=1}^{n} \mathbb{E}_q[\log f(y_i \mid \mathbf{x}_i, \theta)] - \text{KL}(q \,\|\, p)$.
- Explicit observation noise σ^2 in Gaussian regression (`sigma` in `lbbnn_net()`) prevents likelihood mis-scaling ($T = \sigma^2 \neq 1$) and protects spike-and-slab KL shrinkage.

Open-Source Software Ecosystem:

- **R Package `LBBNN`**:
  - Available on CRAN (v0.1.6).
  - Built on `torch` for R (pure C++ LibTorch backend, zero Python overhead).
  - Exported helpers for dataloaders, survey weighting, and custom likelihoods.
- **Python Package `ISLaB-LBBNN`**:
  - PyTorch implementation supporting large-scale GPU cluster distributed training.

Section 2

PART II: Track 1 — Unit-Weighted Pseudo-Posteriors & Formal Differential Privacy

Track 1 Overview — Unit-Weighted Variational Pseudo-Posteriors

BLS Collaboration with Terrance D. Savitsky, Matthew R. Williams & Vladislav Beresovsky:

- Equipping each unit $i \in \{1, \ldots, n\}$ with a non-negative weight $u_i \geq 0$ defines the variational pseudo-posterior (Savitsky & Toth, 2016; Bhattacharya et al., 2019):

$$p_{\mathbf{u}}(\theta \mid \mathbf{y}) \propto p(\theta) \prod_{i=1}^{n} f(y_i \mid \mathbf{x}_i, \theta)^{u_i} \iff \mathsf{ELBO}(q; \mathbf{u}) = \sum_{i=1}^{n} u_i \, \mathbb{E}_{q(\theta)}[\log f(y_i \mid \mathbf{x}_i, \theta)] - \mathsf{KL}(q \parallel$$

Two Distinct Operational Regimes Implemented in `LBBNN`:

1. **Survey Sampling Weights ($u_i = \tilde{w}_i$):**
   - Design weights $w_i = \pi_i^{-1}$ are sample-size normalized: $\tilde{w}_i = w_i \frac{n}{\sum_{j=1}^{n} w_j}$ (`normalize_weights = TRUE`).
   - Enforces $\sum \tilde{w}_i = n$, ensuring the data likelihood does not overpower the spike-and-slab KL penalty.
2. **Differential Privacy Weights ($u_i = \alpha_i \in [0, 1]$):**
   - Risk-downweighting factors are left unnormalized (`normalize_weights = FALSE`, $\sum \alpha_i \leq n$).

Survey-Weighted Imputation Under Informative Sampling (Setup)

True Non-Linear DGP: $Y_i = 2.0X_{1i} + 1.5\sin(\pi X_{2i}) + 0.8X_{1i}X_{2i} + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 0.4^2)$, $X_3, X_4 \sim \mathcal{N}(0,1)$

Simulation Setup ($N = 2,000, p = 4$):

- **Informative Selection Mechanism**:

$$\pi_i = \text{logit}^{-1}(-1.0 + 1.2Y_i)$$

  Draws $n = 737$ units favoring high Y_i.

- Biased sample mean: **+2.009** (True: **0.000**).

Variational Inclusion Probabilities (Table 1):

Variable	Paths	Out Incl.	Status
x1 (Linear)	1	0.999	Selected
x2 (Non-linear)	1	0.999	Selected
x3 (Noise)	0	0.018	Pruned
x4 (Noise)	0	0.130	Pruned

Takeaway: Both true signals X_1, X_2 are retained (Out Incl. $= 0.999$, Paths $= 1$). Pure noise variables X_3, X_4 are cleanly pruned (Paths $= 0$, Out Incl. ≤ 0.130).

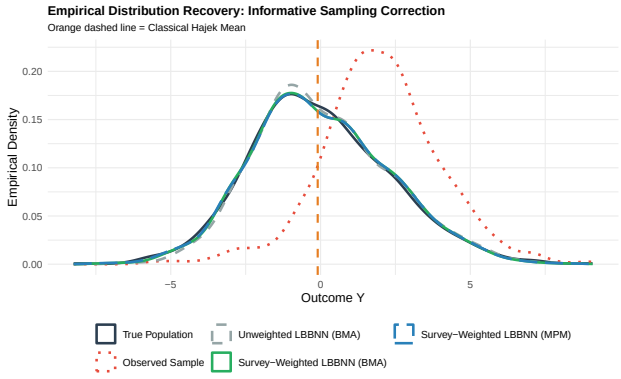
Survey-Weighted Imputation — Population Recovery & Density Plot

Imputation Accuracy (Table 2):

Model	Est	Bias	RMSE
Biased Sample Mean (Naive)	2.009	2.009	3.057
Hajek Weighted Sample Mean	0.271	0.272	2.320
Unweighted LBBNN (BMA)	0.142	0.142	0.541
Unweighted LBBNN (MPM)	0.140	0.140	0.539
Survey-Weighted LBBNN (BMA)	0.039	0.039	0.461
Survey-Weighted LBBNN (MPM)	0.042	0.042	0.461

Key Insights:

- Classical **Hájek mean** reduces bias ($+0.272$) but only estimates a single scalar (RMSE $= 2.320$).
- **Survey-Weighted LBBNN** learns the conditional surface, reducing RMSE to **0.461** (approaching $\sigma = 0.40$ noise floor).



Two-Pass (ϵ, δ)-Differential Privacy (Savitsky et al., 2022)

Why Unweighted Neural Networks Fail DP on Skewed Survey Data:

- Pointwise log-likelihood $|\log f(y_i \mid \mathbf{x}_i, \theta)| \propto \frac{(y_i - \mu_\theta(\mathbf{x}_i))^2}{2\sigma^2}$ grows **quadratically** with the residual.
- In skewed expenditure/income data, extreme outliers drive local sensitivity $\Delta \to \infty$, destroying privacy.

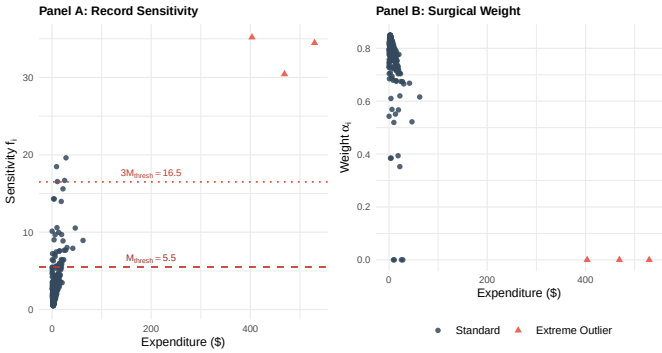
The Two-Pass Surgical Mechanism (`fit_dp_lbbnn`):

1. **Pass 1 (Exploration)**: Fit unweighted LBBNN ($u_i = 1$). Extract $S \times n$ matrix $L^{(1)}_{s,i} = \log f(y_i \mid \mathbf{x}_i, \theta^{(1,s)})$ via `loglik_contributions()`, and evaluate worst-case sensitivity $f_i = \max_s |L^{(1)}_{s,i}|$.
2. **Monotone Surgical Weights**: Compute smooth risk weights $\alpha_i^{\text{raw}} = c_{\text{scale}}(1 - \tilde{f}_i)$, cap at $\alpha_i \leftarrow \min(\alpha_i^{\text{raw}}, M_{\text{thresh}}/f_i)$, and **surgically zero extreme tail outliers**:

$$\alpha_i = 0 \quad \text{if } f_i > 3M_{\text{thresh}}$$

3. **Pass 2 (α-Weighted Fit & Synthesis)**: Fit LBBNN with $u_i = \alpha_i$. Verify local Lipschitz bound $\Delta_{\text{max}} = \max_s |\alpha_i L^{(2)}| \leq M_{\text{thresh}}$, and privacy budget $\epsilon = 2\Delta$

Two-Pass DP Synthesizer — Monotone Surgical Diagnostic



Empirical DP Metrics:

- **Sample**: Skewed log-normal expenditure ($n = 300$, median $\$4.16$, 3 extreme outliers at $\$400$–$\$530$).
- **Standard Records**: Retain high utility weights $\alpha_i \in [0.34, 0.85]$ (median 0.81).
- **Tail Outliers**: All 3 extreme outliers ($f_i > 3M_{\text{thresh}}$) are **surgically zeroed** ($\alpha_i = 0.0$).
- **Formal Privacy Guarantee**:

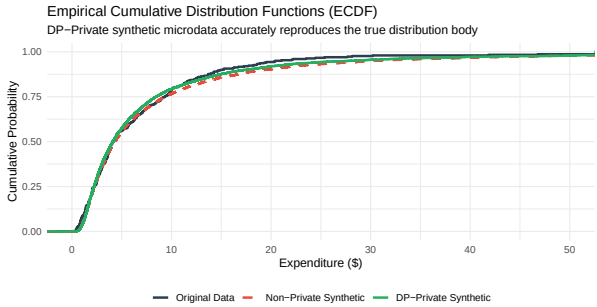
$$\Delta_{\alpha,\mathbf{y}} = 4.32 \leq 5.50 \implies \epsilon_{\mathbf{y}} = 8.63$$

Two-Pass DP Synthesizer — Utility vs. Outlier Protection

Utility Benchmark (Table 3):

Metric	Clean	Outliers	Non-Priv	DP-Priv
Mean	6.70	11.30	7.64	7.99
Std. Dev.	7.23	46.75	20.03	13.95
25th Pct	2.20	2.22	2.00	2.08
Median	4.10	4.16	3.45	4.13
75th Pct	9.09	9.31	6.56	8.72
90th Pct	14.40	14.96	14.14	17.10
Max (Extreme)	62.54	529.57	267.96	156.25

- 3 outliers inflate raw mean from $\$6.70$ to $\$11.30$ and SD from $\$7.23$ to $\$46.75$.
- **DP-Private Synthetic** recovers the true body (Mean $\$7.99$, Median $\$4.13$) while truncating extreme risk.



Section 3

PART III: Track 2 — Combining Probability & Non-Probability Samples

Track 2 Overview — Selection Bias & The Conditioning Trick

Collaboration with Vladislav Beresovsky, Julie Gershunskaya & Terrance D. Savitsky (2026):

- **The Big Data Paradox (Meng, 2018)**: Opt-in convenience samples S_c suffer from severe selection bias:

$$\text{Bias}(\bar{y}_c) \approx \rho_{R,Y} \cdot \sigma_Y \cdot \sqrt{\frac{N - n_c}{n_c}}$$

  Massive convenience sample size n_c merely shrinks variance around an egregiously biased estimate.

The Identification Barrier & The Conditioning Trick:

- **Barrier**: Covariates $\mathbf{x}_i$ are observed only for $i \in S_c$ ($R_i = 1$, unknown π_{ci}) and for a reference probability survey S_r ($I_i = 1$, known design probability π_{ri}), never on $U \setminus (S_c \cup S_r)$.
- **Conditioning Trick**: By pooling S_c and S_r and conditioning on selection into the combined sample S, the known reference design probability π_{ri} acts as a **structural offset** that anchors neural network logits $\eta_\theta(\mathbf{x}_i)$ to the true population scale:

Five Sample Combination Formulations (Beresovsky et al., 2026)

Estimator	Sample Domain	Target Indicator	Composite Probability π_{zi}
`STACK`	$S_c \oplus S_r$ (Unlinked)	$I_{zi} = D_i$ (sample source)	$\frac{\pi_{ci}}{\pi_{ci}+\pi_{ri}}$
`LINK1`	$S_c \cup (S_r \setminus S_c)$	$I_{zi} = I_{ci}$ (deduplicated)	$\frac{\pi_{ci}}{\pi_{ci}+\pi_{ri}-\pi_{ci}\pi_{ri}}$
`LINK2`	$S_c \oplus S_r$ (Linked)	$I_{zi} = D_i + (1-D_i)I_{ci}$	$\frac{\pi_{ci}(1+\pi_{ri})}{\pi_{ci}+\pi_{ri}}$
`MIP`	Missing Information	$\tilde{I}_{zi}^{\text{MIP}} = D_i + (1-D_i)\tilde{p}_i$	Bridges `STACK` ($p=0$) and `LINK2` ($p=1$)
`PSEUDO`	Survey Weighted	Score: $\nabla \ell_{\text{pseudo}} = 0$	Design-weighted: $\sum_{S_r} \pi_{ri}^{-1}\text{softplus}(\eta_i)$

Structural Characteristics:

- **Deterministic Linkage (`LINK1`, `LINK2`)**: Utilizes unit-level linkage overlap between S_c and S_r to maximize Fisher information and minimize asymptotic variance.
- **Missing Information Principle (`MIP`)**: Continuously bridges unlinked `STACK` ($p_{\text{link}} = 0$) and fully linked `LINK2` ($p_{\text{link}} = 1$) via EM expectations.
- **Pseudo-Likelihood (`PSEUDO`)**: Vulnerable to catastrophic weight explosion when reference probabilities $\pi_{ri} \approx 0$.

Study 0 — Exact Linear Baseline Monte Carlo Benchmark ($R = 20$)

High Common Support (Table 2):

Estimator	MC Mean	Bias	SD	RMSE
Naive Convenience Mean	2.095	1.052	0.115	1.058
Oracle Hajek	1.085	0.041	0.200	0.204
GLM-STACK	1.047	0.004	0.234	0.234
GLM-LINK1	1.050	0.007	0.240	0.240
GLM-LINK2	1.054	0.010	0.240	0.240
GLM-MIP50	0.921	-0.123	0.275	0.301
GLM-PSEUDO	1.051	0.007	0.288	0.288
LBBNN-LINK2 (BMA)	0.993	-0.050	0.225	0.231

- **Efficiency Hierarchy**: LINK2 $\succ$ LINK1 $\succ$ STACK $\succ$ PSEUDO.
- **LBBNN Agreement**: Matches GLM efficiency on linear truth (RMSE 0.231 vs. 0.240).

Low Common Support (Table 3):

Estimator	MC Mean	Bias	SD	RMSE
Naive Convenience Mean	-0.777	-1.821	0.113	1.824
Oracle Hajek	1.040	-0.004	0.224	0.224
GLM-STACK	1.024	-0.020	0.133	0.135
GLM-LINK1	1.019	-0.024	0.133	0.136
GLM-LINK2	1.015	-0.029	0.135	0.138
GLM-MIP50	1.450	0.407	0.164	0.438
GLM-PSEUDO	1.541	0.497	0.885	1.015
LBBNN-LINK2 (BMA)	0.934	-0.110	0.128	0.169

- **Pseudo Breakdown**: `GLM-PSEUDO` SD (0.885) and RMSE (1.015) explode under low support.
- **Composite Stability**: `STACK`, `LINK2`, and `LBBNN` bound loss gradients (RMSE 0.13–0.17).

Study 1 — Automated Confounder Selection & Noise Pruning ($p = 6$)

Simulation Setup ($N = 5,000, p = 6$):

- **Confounders**: X_1, X_2 drive both propensity π_{ci} and outcome Y_i.
- **Outcome-Only**: X_3 drives Y_i but not π_{ci}.
- **Pure Noise**: $X_4, X_5, X_6 \sim \mathcal{N}(0, 1)$.

Why Feature Selection Matters:

- In standard GLMs, including outcome-only or noise predictors in propensity models inflates Hájek weight variance without reducing confounding bias.
- `LBBNN` automatically prunes non-propensity predictors.

Variational Edge Probabilities (Table 4):

Variable	Role	Out Incl.	Status
X1 (Active)	Confounder	0.997	Selected
X2 (Active)	Confounder	0.991	Selected
X3 (Outcome)	Outcome Only	0.028	Pruned
X4 (Noise)	Pure Noise	0.026	Pruned
X5 (Noise)	Pure Noise	0.015	Pruned
X6 (Noise)	Pure Noise	0.038	Pruned

- Active confounders X_1, X_2 selected ($a_1 \geq 0.99$).
- X_3 and noise X_4–X_6 completely pruned ($a_1 \leq 0.038$).

Study 2 — High Common Support with Non-Linear Propensity & Interaction

Non-Linear DGP Surface: $\eta_c(\mathbf{x}_i) = \beta_0 + 1.0X_{1i} - 0.5X_{2i} + 1.2X_{1i}X_{2i} + \cos(\pi X_{1i}) - 0.8X_{2i}^2$

Main-Effects GLM Failure (Table 7):

Estimator	Est	Bias	Abs Err
Naive Convenience Mean	3.438	1.919	1.919
Oracle Hajek	1.314	-0.205	0.205
GLM-LINK1	1.131	-0.388	0.388
GLM-LINK2	1.126	-0.393	0.393
LBBNN-STACK (BMA)	1.413	-0.106	0.106
LBBNN-STACK (MPM)	1.433	-0.086	0.086
LBBNN-LINK2 (BMA)	1.214	-0.305	0.305
LBBNN-LINK2 (MPM)	1.222	-0.297	0.297

Key Insights:

- Even under high common support, linear GLMs are **structurally misspecified**:
  - `GLM-LINK1` bias: -**0.388**
  - `GLM-LINK2` bias: -**0.393**
  - `GLM-PSEUDO` bias: -**0.389**
- **LBBNN Non-Linear Recovery**:
  - Hidden layer captures X_1X_2 interaction and trigonometric curvature.
  - `LBBNN-STACK (MPM)` bias: -**0.086** (4.5x reduction!).
  - `LBBNN-LINK2 (MPM)` bias: -**0.297**.
- Prunes noise features X_3–X_6 ($a_1 \leq 0.066$).

Study 3 — Low Common Support + Non-Linearity & Kish ESS

Severe Misspecification + Low Support (Table 8):

Estimator	Est	Bias	Abs Err	Kish ESS
Naive Convenience Mean (Unweighted)	0.064	-1.455	1.455	521.000
Oracle Hajek (True Propensity)	1.480	-0.040	0.040	183.744
GLM-STACK	2.381	0.862	0.862	107.230
GLM-LINK2	2.458	0.939	0.939	99.654
GLM-PSEUDO	6.576	5.057	5.057	6.562
LBBNN-STACK (BMA)	2.376	0.857	0.857	113.702
LBBNN-STACK (MPM)	2.423	0.904	0.904	109.201
LBBNN-LINK2 (BMA)	2.282	0.763	0.763	123.490
LBBNN-LINK2 (MPM)	2.127	0.608	0.608	135.985

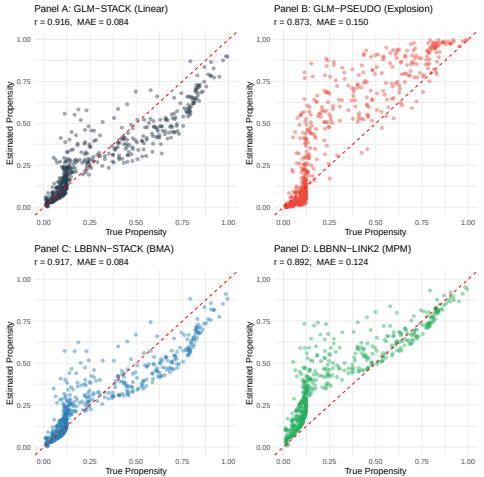
Catastrophic Breakdown of PSEUDO:

- When π_{ci} is high where $\pi_{ri} \approx 0$, pseudo weights $w_i = \pi_{ri}^{-1}$ explode.
- **Kish ESS collapses to 6.56** (out of $n_c = 521$!), causing bias to blow up to **+5.057**.

Composite LBBNN Robustness:

- Relative odds $\pi_{zi} = \frac{\pi_{ci}}{\pi_{ci}+\pi_{ri}}$ bound gradients.
- `LBBNN-LINK2 (MPM)` achieves lowest bias (**+0.608**) and highest effective sample size (**Kish ESS = 135.98**).

Propensity Recovery Diagnostic — 2x2 Comparison Grid



Panels: (A) GLM-STACK bias from missing interactions; (B) GLM-PSEUDO severe weight explosion and scatter collapse; (C) LBBNN-STACK Bayesian Model Averaging calibration; (D) LBBNN-LINK2 Median Probability Model point recovery.

Section 4

PART IV: Unified Software Architecture & Synthesis

Unified LBBNN Architecture Across Track 1 & Track 2

Function / Module	Operational Track	Description & Statistical Role
`get_dataloaders()`	Track 1 & 2	Handles non-trainable weights $u_i \geq 0$, normalization, and multi-targets.
`lbbnn_net(sigma, ...)`	Track 1 & 2	ELBO likelihood scaling by error variance σ^2 (cold-posterior protection).
`loglik_contributions()`	Track 1B (Privacy)	Computes $S \times n$ record-level log-likelihood contributions $L_{s,i}$.
`fit_dp_lbbnn()`	Track 1B (Privacy)	Two-pass monotone surgical downweighting with Lipschitz bound Δ_α.
`sample_synthetic_lbbnn()`	Track 1B (Privacy)	Generates (ϵ, δ)-DP synthetic microdata via post-processing theorem.
`nll_link1()` / `nll_link2()`	Track 2 (Propensity)	Composite Bernoulli NLL on singly matched and matched-pair samples.
`make_nll_mip(p_link)`	Track 2 (Propensity)	Factory producing composite NLL under Missing Information Principle.
`nll_stack()` / `nll_pseudo()`	Track 2 (Propensity)	Stacked pooled sample NLL and survey design-weighted pseudo-NLL.
`build_propensity_dataset()`	Track 2 (Propensity)	Formats covariates and targets for any of the 5 estimation methods.
`hajek_estimate()`	Track 2 (Propensity)	Computes self-normalizing Hájek estimator $\sum w_i y_i / \sum w_i$ with $w_i = \hat{\pi}_{ci}^{-1}$.

Key Takeaways & Synthesis for Official Statistics

1. **Survey-Weighted Pseudo-Posteriors (Track 1A)**:
   - Normalizing survey weights to sample size ($\sum \tilde{w}_i = n$) + calibrated observation noise σ^2 prevents likelihood mis-scaling ($T \neq 1$) and protects spike-and-slab KL shrinkage.
   - `LBBNN` achieves 5x lower population RMSE than classical Hájek weighting under informative sampling while pruning irrelevant covariates.
2. **Two-Pass (ϵ, δ)-PDP Microdata Synthesis (Track 1B)**:
   - Monotone surgical downweighting (`fit_dp_lbbnn`) eliminates disclosure risk from extreme tail outliers ($\alpha_i = 0$) while preserving high utility ($\alpha_i \in [0.34, 0.85]$) on normal records.
3. **Composite Non-Probability Sample Integration (Track 2)**:
   - Conditioning on pooled sample inclusion $S_c \cup S_r$ allows known design probabilities π_{ri} to anchor neural network logits to the true population scale.
   - Composite Bernoulli likelihoods (`LINK2`, `LINK1`, `MIP`, `STACK`) bound loss gradients and prevent the catastrophic effective sample size collapse (`Kish ESS` < 7) of pseudo-likelihood under low common support.

References & Collaborative Works

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